

MD. MIFTAHUR RAHMAN SWAPNIL

Product Designer & Builder | Turning Real-World Problems into Digital Products

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SUMMARY

Final-year CSE undergraduate and product-oriented builder with hands-on experience turning real-world problems into digital products across UI/UX design, modern frontend engineering (Next.js, TypeScript), and AI system integrations. Creator of Edu51Portal, currently serving ~100 active university students with \$0 cloud infrastructure bills. Undergraduate researcher with accepted international publication at IEEE OMLET 2026. Proven ability to bridge product prototyping, high-fidelity design systems, and resilient production code.

SKILLS

- Product & Design:** UI/UX Design, Digital Product Design, Wireframing, High-Fidelity Prototyping (Figma), Design Systems, User Research
- Frontend Engineering:** Next.js 14/15 (App Router), React, TypeScript, Tailwind CSS, JavaScript (ES6+), HTML5/CSS3, Responsive UI
- Backend & Database:** Supabase (PostgreSQL, Row Level Security, Auth), Node.js, REST APIs, Google Drive API, Firebase
- AI & Research:** LLM Prompting & RAG Systems, Relational Graph Convolutional Networks (R-GCN), Neo4j, PyTorch, Python
- Tools & Automation:** Git & GitHub, n8n Automation, Vercel, Docker (Basic), VS Code, LaTeX, Overleaf

PROJECTS

- Edu51Portal** — Next.js 14, TypeScript, Supabase PostgreSQL, Google Drive API, Tailwind CSS Live
- Designed and built an academic management platform serving **~100 active BUBT students**, delivering sub-second routine lookups, notices, and course handouts.
 - Integrated Google Drive API with Supabase to distribute 1,000+ academic PDFs securely with **\$0 cloud storage infrastructure overhead**.
 - Implemented section-specific schedule filtering, fuzzy course search, and mobile-first responsive architecture.
- CurricuRAG** — PyTorch, Relational GCN, Neo4j, Qwen2.5-7B-Instruct, Sentence-BERT Accepted (IEEE OMLET 2026)
- Engineered a curriculum knowledge graph RAG system with a 2-layer R-GCN encoder and DistMult decoder over **418 nodes and 558 typed edges** verified in Neo4j.
 - Achieved **45.5% exact-set match** (vs. 22.7% text-RAG) with a **100% correct abstention rate** on unanswerable questions and 0 query-time LLM overhead.
- OpusGen AI** — Next.js, TypeScript, Generative Diffusion Models, REST APIs, Vercel Live
- Built an AI creative marketing suite providing automated studio lighting, background replacement, and 4x neural upscaling for e-commerce merchant catalogs.
 - Constructed batch processing export pipelines and multi-ratio ad canvas expansion, accelerating creative workflows from hours to under 30 seconds.
- Smart Classroom Safety Automation** — ESP32 DevKit, ESP32-CAM, Firebase, Expo Android, Edge AI Completed
- Engineered an IoT smart classroom safety and environmental monitoring system integrating sensor telemetry (DHT11, PIR, Ultrasonic, Sound) with automated door actuators and alert buzzers.
- Stark-OS Command Portfolio** — Next.js 14, TypeScript, Tailwind CSS, Supabase Telemetry, Vercel Live
- Engineered an interactive Iron Man Stark-OS personal portfolio (mrswapnil.me) with HUD audio feedback, multi-theme presets, and 98+ Lighthouse performance score.

RESEARCH & PUBLICATIONS

- Relation-Aware Graph Retrieval over a Curriculum Knowledge Graph for Prerequisite QA** Oct 2026
- IEEE OMLET 2026** (Nairobi, Kenya) — Accepted with Minor Revision (Paper ID: 1017, Scheduled for IEEE Xplore & Scopus).
 - Authors: Md. Jahidul Kamal Islam, **Md. Miftahur Rahman Swapnil**, Md. Asif Ali, Shrabani Das, Shefayatuj Johara Chowdhury (BUBT CSE).

EDUCATION

- Bachelor of Science in Computer Science & Engineering (CSE)** CGPA: 3.60 / 4.00
Bangladesh University of Business and Technology (BUBT) | Intake 51 2022 – 2026 (Expected)
- Higher Secondary Certificate (HSC) — Science** GPA: 5.00 / 5.00
Shaheed Police Smrity College 2021
- Secondary School Certificate (SSC) — Science** GPA: 5.00 / 5.00
Shaheed Police Smrity College 2019

LEADERSHIP & VOLUNTEERING

- BUBT IT Club & CSE Department Community** — Active Contributor & Technical Mentor 2022 – Present
- Founder & Lead Maintainer — Edu51Portal** — Continuous support and tooling for ~100 BUBT peers 2023 – Present

LANGUAGES

- Bangla** — Native / First Language
- English** — Professional Working Proficiency